

MAT120- Finals

$$L\{y'' + p(x)y' + q(x)y\} = \frac{b}{s^2} \cdot \frac{1}{s^2}$$

Final Topics:-

⇒ Separable Variable

⇒ Linear First Order ODE

⇒ Exact - Non exact DE

⇒ Homogeneous - Non homogeneous DE

⇒ Application of First Order ODE

⇒ Bernoulli's Equation

⇒ Double Integral

⇒ ~~Triple Integral~~

Multivariable

* Introduction to Differential Eqn *

⇒ A differential equation is any eqn which contains derivatives either ordinary or partial derivatives.

I ODE [Ordinary DE]

II PDE [Partial DE]

III DAE [Differential Algebraic Eqn]

$$0 = y'' + \frac{y^2}{x^2} + \frac{y}{x^2}$$

Example

$$\frac{d}{dx}, \frac{d}{dy} = \boxed{\text{ }} \Rightarrow \text{ODE}$$

$$\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} = \boxed{\text{ }} \Rightarrow \text{PDE}$$

DE

* ODE:

$$a \left[\frac{d^2 y}{dx^2} \right] + b \frac{dy}{dx} + cy = f(x)$$

* PDE:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

* Order of DE:-

The order of Any DE is the largest derivative present in the DE.

Example:

$$y''' + 2y'' + 3y' + 4y = 0$$

$$\left[\frac{d^3 y}{dx^3} \right] + 2 \frac{d^2 y}{dx^2} + 3 \left[\frac{dy}{dx} \right] + 4y = 0$$

* Degree of DE:

This is the degree of the Order of DE.

$$3^{\text{rd}} \text{ Order} \leftarrow \left(\frac{d^3 y}{dx^3} \right) + 2 \frac{d^2 y}{dx^2} + 4y = 0 \rightarrow 1^{\text{st}} \text{ Degree}$$

Exp $\left(\frac{d^2y}{dx^2}\right)^2 + 2\frac{dy}{dx} + 4y = 0$

$\Rightarrow$ Order : 2

$\Rightarrow$ Degree : 2

Note : $\left(\frac{d^2y}{dx^2}\right)^2 \neq \frac{d^4y^2}{dx^4}$

Exp

$\frac{d^3y}{dx^3}$

$\Rightarrow$ Order : 03 Degree : 1

Exp

$\frac{d^2y}{dx^2} = \sqrt{\left(\frac{dy}{dx}\right)^2 + 3y}$

$\Rightarrow \left(\frac{d^2y}{dx^2}\right)^2 = \left(\frac{dy}{dx}\right)^2 + 3y$

$\therefore$ Order : 2 , Degree : 2

Exp

$1 + \left(\frac{dy}{dx}\right)^4 = \frac{d^3y}{dx^3}$

Order : 3 , Degree : 1

* Classification of ODE :

$\Rightarrow$ Classification By Order

$\Rightarrow$ Classification By Linearity

$\hookrightarrow$ Linear DE

$\hookrightarrow$ Non Linear DE

Identifying Non Linear DE :-

- ⇒ Dependent Variable $\times$ Dependent Variable
- ⇒ Dependent Variable $\times$ Derivative of Dep. Variable
- ⇒ Derivative of D.V $\times$ Derivative of D.V
- ⇒ Transcendental function of Dep. Variable

EXP $y^2 \cdot \frac{dy}{dx} + 2y = 0$ 1st Order 1st Degree

↳ DV $\times$ DV $\Rightarrow$ Non Linear

EXP $y \frac{dy}{dx} + 2y = 0$ 1st Order 1st degree

↳ D.V $\times$ D.D.V $\Rightarrow$ Non Linear

EXP $\left(\frac{dy}{dx}\right)^2 + 2y = 0$ 1st Order 2nd degree

⇒ $\frac{dy}{dx} \cdot \frac{dy}{dx} + 2y = 0$

↳ D.D.V $\times$ D.D.V $\Rightarrow$ Non Linear

* Transcendental : Applies to Dep. Variable

$$f(x) = \ln x / \ln y$$

$$f(x) = e^x / e^y$$

$$f(x) = \sin x / \sin y$$

$$f(x) = \text{Polynomial}$$

Exp

$$1. \frac{dy}{dx} + 2y = \sin x \rightarrow \text{Linear [Dep } y=x]$$

$$\frac{dy}{dx} + 2y = \sin y \rightarrow \text{Non Linear}$$

$$\frac{d^2x}{dy^2} + 2 \frac{dx}{dy} = 4y^3 \rightarrow \text{Linear [Dep } y=x]$$

Lec-02
RHR

Method of Separation of Variables [1st Order]

$\Rightarrow$ Separable Equation:

A 1st order DE of the form $\frac{dy}{dx} = g(x) \cdot h(y)$ is said to be separable equation.

$$\frac{dy}{h(y)} = g(x) dx + C = \int g(x) dx + C$$

$$\Rightarrow \int \frac{1}{h(y)} dy = \int g(x) dx$$

$\therefore y = \underline{\hspace{2cm}}$ [Solution]

Exp-01

Solve the DE, $y(0) = 1$

$$y \frac{dy}{dx} = 2x(y+1)$$

$$\Rightarrow y \frac{dy}{dx} = 2x(y+1)$$

$$\Rightarrow \frac{y}{y+1} dy = 2x dx$$

$$\Rightarrow \int \frac{y}{y+1} dy = \int 2x dx$$

$$\Rightarrow \int \frac{y+1-1}{y+1} dy = 2 \int x dx$$

$$\Rightarrow \int 1 - \frac{1}{y+1} dy = 2 \cdot \frac{x^2}{2}$$

$$\Rightarrow y - \ln|y+1| = x^2 + c$$

if $x=0, y=1$

$$\therefore 1 - \ln|1+1| = 0 + c \Rightarrow c = 1 - \ln 2$$

$$\therefore c = 1 - \ln 2$$

$$\therefore y - \ln|y+1| = x^2 + 1 - \ln 2$$

Exp-2

$2xy' = 1+y^2$, $y(2) = 3$ Solve the DE

$$\Rightarrow 2xy' = 1+y^2$$

$$\Rightarrow 2x \frac{dy}{dx} = 1+y^2$$

$$\Rightarrow 2x \frac{1}{dx} = (1+y^2) \cdot \frac{1}{dy}$$

$$\Rightarrow \frac{1}{2x} dx = \frac{1}{1+y^2} dy$$

$$\Rightarrow \frac{1}{2} \int \frac{1}{x} dx = \int \frac{1}{1+y^2} dy$$

$$\Rightarrow \frac{1}{2} \ln|x| = \ln|1+y^2| + C \quad [\text{General Solution}]$$

if $x=2$, $y=3$

$$\Rightarrow \frac{1}{2} \ln|2| = \ln|1+3^2| + C$$

$C = -1.956$

$$\Rightarrow \frac{1}{2} \ln|x| = \ln|1+y^2| - 1.956 \quad [\text{Particular Solution}]$$

* n^{th} Order Ordinary differential Equation

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x) y = f(x)$$

⇒ Linear 1st Order ODEs

Of The Form : $a_1(x) \frac{dy}{dx} + a_0(x)y = f(x)$

is said to be linear eqn in the variable y .

Standard Form :

$$\frac{dy}{dx} + \frac{a_0(x)}{a_1(x)} y = \frac{f(x)}{a_1(x)}$$

$$\Rightarrow \frac{dy}{dx} + p(x)y = Q(x)$$

$$\frac{a_0(x)}{a_1(x)} = p(x) \quad \left| \quad \frac{f(x)}{a_1(x)} = Q(x) \right.$$

⇒ Steps To Solve :

⇒ Put the linear equation in the standard form.

⇒ Identify $p(x)$ and find the Integrating factor

IF of the form, where $IF = e^{\int p(x) dx}$

⇒ Multiply both sides by IF.

⇒ The L-Hand Side will be $\frac{d}{dx} [IF \cdot y] = Q(x) \cdot IF$

⇒ Integrate both sides and solve for y on D.V

Exp-01

$$x \frac{dy}{dx} - 4y = x^6 e^x$$

Soln:

$$x \frac{dy}{dx} - 4y = x^6 e^x$$

$$\Rightarrow \frac{dy}{dx} - \frac{4}{x}y = x^5 e^x$$

$$\Rightarrow \frac{dy}{dx} + \left(-\frac{4}{x}\right)y = x^5 e^x \quad \text{[Standard form]}$$

Here,

$$P(x) = \left(-\frac{4}{x}\right)$$

$$Q(x) = x^5 e^x$$

Now,

$$IF = e^{\int P(x) dx}$$

$$= e^{\int -\frac{4}{x} dx}$$

$$= e^{-4 \int \frac{1}{x} dx}$$

$$= e^{-4 \ln|x|}$$

$$= e^{\ln x^{-4}}$$

$$IF = \frac{1}{x^4}$$

$$\Rightarrow \frac{1}{x^4} \left[\frac{dy}{dx} + \left(-\frac{4}{x}\right)y \right] = \frac{1}{x^4} x^5 e^x$$

$$\Rightarrow \frac{d}{dx} \left[\frac{1}{x^4} \cdot y \right] = x e^x$$
$$\Rightarrow \int d \left[\frac{y}{x^4} \right] = \int x e^x dx$$

$$\Rightarrow \frac{y}{x^4} = x \int e^x dx - \int \left[\frac{d}{dx}(x) \int e^x dx \right] dx$$

$$= x e^x - \int 1 \cdot e^x dx$$

$$\frac{y}{x^4} = x e^x - e^x$$

$$\Rightarrow y = x^4 (x e^x - e^x) + C$$

Exp-2

$$\frac{dy}{dt} + \frac{y}{t} = e^t$$

Soln:

$$\frac{dy}{dt} + \frac{1}{t} y = e^t$$

$$\Rightarrow P(t) = \frac{1}{t} \quad Q(t) = e^t$$

$$\therefore IF = e^{\int P(t) dt}$$

$$= e^{\int \frac{1}{t} dt}$$

$$= e^{\ln t}$$

$$= t$$

$$t \left[\frac{dy}{dt} + \frac{1}{t} y \right] = t [e^t]$$

$$\Rightarrow \frac{d}{dt} [t \cdot y] = t e^t$$

$$\Rightarrow \int d[t \cdot y] = \int t e^t dt$$

$$\Rightarrow t \cdot y = t \int e^t dt - \int \left[\frac{d}{dt} t \int e^t dt \right] dt$$

$$t \cdot y = t e^t - e^t$$

$$y = \frac{t e^t - e^t}{t} + C \text{ (Soln)}$$

Exp-3

$$y' + y \tan x = \sin 2x$$

Soln:

$$y' + y \tan x = \sin 2x$$

$$\Rightarrow \frac{dy}{dx} + y \tan x = \sin 2x$$

$$P(x) = \tan x, Q(x) = \sin 2x$$

$$IF = e^{\int P(x) dx}$$

$$= e^{\int \tan x dx}$$

$$= e^{\ln |\sec x|}$$

$$= \sec x$$

$$\therefore \sec x \left[\frac{dy}{dx} + y \tan x \right] = \sec x [\sin 2x]$$

$$\Rightarrow \int \frac{d}{dx} [\sec x \cdot y] = \sin 2x \sec x$$

$$\Rightarrow \int d(y \sec x) = \int \frac{2 \sin x \cos x}{\cos x}$$

$$\therefore y \sec x = -2 \cos x + C \quad (\text{Ans})$$

$$Zill \quad 2.2 \Rightarrow 1-18, 23-26, 31, 32$$

$$2.3 \Rightarrow 1-36$$

Lee-03
RHR

* Application of 1st order ODE:-

Exp-01

A culture initially has P_0 number of bacteria. At $t=1$ hr the number of bacteria is measured to be $\frac{3}{2}P_0$. If the rate of growth is proportional to the number of bacteria $P(t)$, present at time t , determine time necessary for the number of bacteria to triple.

Soln:

Given,

$$t=0$$

$$P = P_0$$

$$t=1$$

$$P = \frac{3}{2}P_0$$

We know,

$$\frac{dP}{dt} \propto P$$

$$\Rightarrow \frac{dP}{dt} = KP$$

$$\Rightarrow \int \frac{dP}{P} = \int K dt$$

$$\Rightarrow \ln P = Kt + \ln C$$

$$\Rightarrow \ln P - \ln C - Kt = 0$$

$$\Rightarrow \ln \left(\frac{P}{C} \right) = Kt$$

$$\Rightarrow \frac{P}{C} = e^{Kt}$$

$P = Ce^{Kt}$

Now,

$$t=0, p=p_0$$

$$\textcircled{i} \Rightarrow p_0 = p_0 e^{k \times 0} = p_0$$

$$\textcircled{i} \Rightarrow p = p_0 e^{kt}$$

$$t=1, p = \frac{3}{2} p_0$$

$$\therefore \frac{3}{2} p_0 = p_0 e^{k \times 1}$$

$$\Rightarrow \frac{3}{2} = e^k \quad \leftarrow 0 < k$$

$$\Rightarrow \frac{3}{2} = e^k \quad \leftarrow 0 < k$$

$$\Rightarrow k = \ln\left(\frac{3}{2}\right) = 0.405$$

$$\therefore p = p_0 e^{0.405 t} \quad \textcircled{ii}$$

Now,

$$\text{If } t=2, p=3p_0$$

$$3p_0 = p_0 e^{0.405 \times 2}$$

$$\therefore 3 = e^{0.81}$$

$$\Rightarrow 3 = e^{0.81}$$

$$\Rightarrow 3 = e^{0.81}$$

$$\therefore t = 2.7126 \text{ (hr)}$$

Ans:

$$m \frac{dT}{dt} = -k(T - T_m)$$

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Syllabus :

- ⇒ Growth and Decay Model
- ⇒ Newton's Law of cooling
- ⇒ Circuit

* Growth and Decay Models:

(i) $\frac{dP}{dt} \propto P$

⇒ $\frac{dP}{dt} = KP \rightarrow \begin{matrix} K > 0 \rightarrow \text{Growth} \\ K < 0 \rightarrow \text{Decay} \end{matrix}$

(ii) $P = Ae^{Kt}$

* Newton's Law of cooling:

$\frac{dT}{dt} \propto (T - T_m)$

$\frac{dT}{dt} = K(T - T_m)$

⇒ $\int \frac{dT}{T - T_m} = \int K dt$

⇒ $\ln(T - T_m) = Kt + \ln e$

⇒ $\ln(T - T_m) - \ln e = Kt$

⇒ $\ln\left(\frac{T - T_m}{e}\right) = Kt$

⇒ $\frac{T - T_m}{e} = e^{Kt}$

Now, $T = \text{Body Temperature}$
 $T_m = \text{Surrounding Temp.}$

$T - T_m = e^{Kt}$
 $\therefore \boxed{T = T_m + e^{Kt}}$

Exp-2

The population of a town grows at a rate proportional to the population present at time 't'.

The initial population of 500 increases by 15% in 10 years. What will be the population in 30 years? How fast is the population growing at $t = 30$?

Soln:

Given,

$$t=0 \quad P_0 = 500$$

$$t=10 \quad P_0 = 575 \quad [1.15 P_0]$$

We know,

$$\frac{dP}{dt} \propto P$$

$$\Rightarrow P = P_0 e^{kt}$$

$$\Rightarrow 1.15 P_0 = P_0 e^{k \times 10} \quad | \quad e^{10k} = 1.15$$

$$\Rightarrow \ln(1.15) = 10k$$

$$\therefore k = 0.01398$$

$$\therefore t = 30 \Rightarrow$$

$$P = 500 \times e^{0.01398 \times 30}$$
$$= 761$$

Now,

$$P_{(30)} = P_0 e^{30k}$$

$$P_{(30)} = P_0 (e^{10k})^3$$
$$= P_0 (1.15)^3$$

Now,

$$\frac{dP}{dt} = K \cdot P_{90}$$

$$= 0.01398 \times (1.521 P_0)$$

$$= 0.0213 P_0$$

$\therefore 10.65 \approx 11$ person per year!

Exp-3

When a cake is removed from an oven, its temperature is measured at 300°F . 3 min later, its temperature is 200°F . Find the temperature profile $T(t)$ when the room temperature is 70°F .

Soln: Given,

$$t=0, T=300^\circ\text{F}$$

$$t=3, T=200^\circ\text{F}$$

$$T_m = 70^\circ\text{F}$$

We know,

$$T = T_m + Ce^{Kt}$$

$$= 70 + Ce^{Kt}$$

When,

$$t=0$$

$$300 = 70 + Ce^{Kt}$$

$$\Rightarrow C = 230^\circ\text{F}$$

When $t = 3$,

$$200 = 70 + 230e^{K \times 3}$$

$$\Rightarrow 200 = 70 + 230e^{3K}$$

$$\Rightarrow 130 = 230e^{3K}$$

$$\Rightarrow \ln\left(\frac{130}{230}\right) = 3K$$

$$\therefore K = \frac{\ln\left(\frac{130}{230}\right)}{3} = -0.19$$

$$\therefore T = 70 + 230e^{-0.19 \times t}$$

$$\Rightarrow 1200 = 70 + 230e^{-0.19 \times t}$$

$$\therefore t = 10.72 \text{ (min)}$$

Ans:

When, $t = 10.72$ min

$t = 10.72$

Let a model that will predict the amount of

protein at any time t

What is the half life of the protein?

What is the amount remaining in 200 years?

$$② \quad X = 20 e^{kt}$$

$$X = 20 e^{\frac{1}{10} \ln\left(\frac{19.88}{20}\right) \cdot t}$$

$$@ t = 900$$

$$X = 20 e^{\frac{900}{10} \ln\left(\frac{19.88}{20}\right)}$$

$$= 11.63 \text{ gm (Ans.)}$$

Lec-04

RHR

* Exact Differential equation:

If $F(x, y)$ is a function of x, y then the total differential eqn is $dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy$. [It means that how F changes when both x, y changes a little.]

A 1st Order ODE of the form:

$$\Rightarrow \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = 0$$

$$\Rightarrow M dx + N dy = 0 \quad \text{--- (i)}$$

① is called Exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

then general solution: $\int M dx + \int N dy = C$

$$F(x, y) = \int M dx + \int N \text{ (not containing the } x \text{ term)} dy + C$$

* Linear 1st Order $\rightarrow \frac{dy}{dx} + P(x)y = Q(x)$
 Separable Variable $\rightarrow \frac{dy}{dx} = f(x)g(y)$

ଉଦାହରଣ,

$$M dx = -N dy \text{ ରୂପେ}$$

$$\Rightarrow -N \frac{dy}{dx} = M \Rightarrow \frac{dy}{dx} = -\frac{M}{N}$$

$$\Rightarrow \frac{dy}{dx} + \left(\frac{M}{N}\right)y = 0$$

Q1 Solve the ODE $2xy dx + (x^2 - 1) dy = 0$ — (1)

ସମାଧାନ:

$$M = 2xy, N = (x^2 - 1)$$

$$\frac{\partial M}{\partial y} = 2x, \frac{\partial N}{\partial x} = 2x$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ [Exact]}$$

$$F = \int M dx + \int N \text{ (not containing } x \text{ terms)} dy$$

$$= \int 2xy dx + \int -1 dy$$

$$= 2y \frac{x^2}{2} - y + C \text{ (Ans.)}$$

Q1: Solve the ODE: $(3x^2y - 6x)dx + (x^3 + 2y)dy = 0$

Soln: $M = 3x^2y - 6x$ $N = x^3 + 2y$

$$\frac{\partial M}{\partial y} = 3x^2 \quad \frac{\partial N}{\partial x} = 3x^2$$

Exact

$$F = \int M dx + \int N dy$$

$$= \int (3x^2y - 6x) dx + \int 2y dy$$

$$= \frac{3x^3y}{3} - \frac{6x^2}{2} + \frac{2y^2}{2} + C$$

$$= x^3y - 3x^2 + y^2 + C \quad (\text{Ans})$$

Q1 Solve the DE: $(2x + y - 1)dy + (2y - x - 5)dx = 0$

Soln: $M = 2y - x - 5$ $N = 2x + y - 1$

$$\frac{\partial M}{\partial y} = 2 \quad \frac{\partial N}{\partial x} = 2$$

Exact

$$F = \int M dx + \int N dy$$

$$= \int (2y - x - 5) dx + \int (2x + y - 1) dy$$

$$= 2xy - \frac{x^2}{2} - 5x + \frac{y^2}{2} + y + C$$

* Non Exact DE: $(M dx + N dy) \neq 0$

$$M dx + N dy = 0 \quad \text{--- (i)}$$

If $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} = \frac{M_y}{N_x}$ $x = \frac{M_y}{N_x}$

then eqn is called Non exact DE.

Eqn can be made exact By these steps:

$\Rightarrow$ If, $\frac{(x \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x})}{N} = f(x)$ [Only x terms]

then $IF = e^{\int f(x) dx}$

$\Rightarrow$ If, $\frac{(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y})}{M} = f(y)$ [Only y terms]

then $IF = e^{\int f(y) dy}$

$\Rightarrow$ Then Multiply the DE by IF, then eqn is converted to exact DE and can be solved by previous manner.

$$M_y = \frac{M_y}{N_x}$$

$$N_x = \frac{M_y}{N_x}$$

$$N_x =$$

Exp $xy \, dx + (2x^2 + 3y^2 - 20) \, dy = 0$ ——— ①

$\Rightarrow M = xy$ ——— $N = 2x^2 + 3y^2 - 20$

$\frac{\partial M}{\partial y} = x$ $\frac{\partial N}{\partial x} = 4x$

Non exact

① $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{x - 4x}{2x^2 + 3y^2 - 20} \neq \frac{M f(x)}{N}$

② $\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xy} = \frac{4x - x}{xy} = \frac{3}{y}$

$IF = e^{\int \frac{3}{y} dy} = e^{3 \ln y} = y^3$

$\Rightarrow$ ① $\times IF$

$xy^4 \, dx + (2x^2y^3 + 3y^5 - 20y^3) \, dy = 0$

Now, $M = xy^4$ $N = 2x^2y^3 + 3y^5 - 20y^3$

$\frac{\partial M}{\partial y} = 4xy^3$ $\frac{\partial N}{\partial x} = 4xy^3$

$= 4xy^3$

Exact

$$\therefore F = \int M dx + \int N dy \quad \text{if } M = \dots$$

$$= \int xy^4 dx + \int (-3y^5 - 20y^3) dy$$

$$= y^4 \frac{x^2}{2} + \frac{3y^6}{6} - \frac{20y^4}{4} + C \quad (\text{Ans})$$

Exp

$$(x^2+y^2)dx - xy dy = 0 \quad \text{if } x^2+y^2 = 1$$

$$\Rightarrow M = x^2+y^2 \quad N = -xy$$

$$\frac{\partial M}{\partial y} = 2y \quad \frac{\partial N}{\partial x} = -y$$

Non exact

$$\textcircled{1} \quad \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y+y}{-xy} = -\frac{3}{x} = f(x)$$

$$\text{IF} = e^{\int f(x) dx} = e^{-3 \ln x} = x^{-3} = \frac{1}{x^3}$$

$$\textcircled{11} \quad \frac{1}{x^3} (x^2+y^2) dx - (xy) \frac{1}{x^3} dy = 0$$

$$\Rightarrow x \left(\frac{1}{x} + \frac{y^2}{x^3} \right) dx - \frac{y}{x^2} dy = 0$$

$$\Rightarrow \frac{x^2+y^2}{x^3} dx - \frac{y}{x^2} dy = 0$$

$$M = \frac{x^2+y^2}{x^3}$$

$$\frac{\partial M}{\partial y} = \frac{2y}{x^3}$$

$$N = -\frac{y}{x^2} = -yx^{-2}$$

$$\frac{\partial N}{\partial x} = 2x^{-3}y = \frac{2y}{x^3}$$

$$\therefore -F = \int M dx + \int N y dy + \text{const } M = F$$

$$= \int \frac{x^2+y^2}{x^3} dx + \int y^2 dy + \text{const } M =$$

$$(\Rightarrow) \int \frac{1}{x} dx + \int y^2 dx + \frac{y^2}{2} + \frac{C}{2} M =$$

$$= \ln|x| + y^2 x + \frac{y^2}{2} + C \quad (\text{Ans})$$

Exp

$$(x+4y^3) dy - y dx = 0 \quad \text{---} \quad y^2 + x^2 = M$$

$$y = \frac{MG}{xG} \quad N = x+4y^3 \quad y^2 = \frac{MG}{xG}$$

$$\Rightarrow M = -y$$

$$\frac{\partial M}{\partial y} = -1 \quad \frac{\partial N}{\partial x} = 1$$

$$\text{Non exact} \quad \frac{MG}{xG} - \frac{MG}{xG} = 0$$

$$\textcircled{1} \quad \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = \frac{1+1}{-y} = -\frac{2}{y} \Rightarrow f(y) = -\frac{2}{y}$$

$$\therefore IF = e^{\int f(y) dy} = e^{-2 \int \frac{1}{y} dy} = e^{-2 \ln y} = e^{\ln y^{-2}} = \frac{1}{y^2}$$

Now, IF x $\textcircled{1}$ $\frac{MG}{xG}$

$$\frac{MG}{xG} = \frac{MG}{xG} = M$$

$$\frac{MG}{xG} = \frac{MG}{xG} = M$$

$$\frac{MG}{xG} = M$$

$$\frac{MG}{xG} = \frac{MG}{xG}$$

$$\frac{1}{y^2} (x + 4y^3) dy - y \left(\frac{1}{y^2} \right) dx$$

$$\Rightarrow \frac{x + 4y^3}{y^2} dy - \frac{1}{y} dx$$

$$M = -\frac{1}{y}$$

$$\frac{\partial M}{\partial y} = -y^{-1} = -y^{-2} = \frac{1}{y^2}$$

$$\frac{\partial N}{\partial x} = \frac{1}{y^2}$$

exact

$$\therefore F = \int M dx + \int N dy$$

$$= \int \frac{1}{y} dx + \int \left(\frac{1}{y^2} + 4y \right) dy = V$$

$$= -\frac{1}{y} x + \frac{y^{-2+1}}{-2+1} + 4 \frac{y^{2+1}}{2} + C = \frac{Vb}{xb}$$

$$\frac{Vb}{xb} \cdot \frac{1}{a} = \frac{Vb}{xb} \cdot \frac{1}{(a-1)}$$

We convert it into linear equation by using

transformation

$$V(x) = V(x)q + \frac{Vb}{xb} \cdot \frac{1}{(a-1)}$$

$$V(x) (a-1) = V(x)q (a-1) + \frac{Vb}{xb}$$

$$(iii) \quad V'(x) = V(x)q + \frac{Vb}{xb}$$

Lec-05
RHR

⇒ Bernoulli's Equation / DE

$$\frac{dy}{dx} + p(x)y = Q(x)y^n \quad \text{--- (i)}$$

$$\Rightarrow \frac{1}{y^n} \frac{dy}{dx} + p(x) \frac{y}{y^n} = Q(x)$$

$$\Rightarrow \frac{1}{y^n} \frac{dy}{dx} + p(x) y^{1-n} = Q(x) \quad \text{--- (ii)}$$

Let,

$$v = y^{1-n}$$

$$\frac{dv}{dx} = (1-n) y^{1-n-1} \frac{dy}{dx} + x \frac{1}{y} =$$

$$\Rightarrow \frac{1}{(1-n)} \frac{dv}{dx} = \frac{1}{y^n} \frac{dy}{dx}$$

We convert it into a linear equation by using transformation.

$$\therefore \frac{1}{(1-n)} \frac{dv}{dx} + p(x)v = Q(x)$$

$$\Rightarrow \frac{dv}{dx} + (1-n)p(x)v = (1-n)Q(x)$$

$$\Rightarrow \frac{dv}{dx} + p'(x)v = Q'(x) \quad \text{--- (iii)}$$

$$IF = e^{\int p'(x) dx}$$

⇒ Multiply Both sides with IF and then regular process.

Exp Solve the DE : $\frac{dy}{dx} + y = xy^2$

Soln: $\frac{dy}{dx} + 1 \cdot y = xy^2$ ——— ①

$$\Rightarrow \left[\frac{1}{y^2} + \frac{y}{y^2} \right] = x$$

$$\Rightarrow \frac{1}{y^2} \left(\frac{dy}{dx} + y \right) = x$$
 ——— ②

Let,

$$v = \frac{1}{y^2} \Rightarrow \frac{dv}{dx} = -\frac{1}{y^3} \frac{dy}{dx}$$

$$\frac{dv}{dx} = -\frac{1}{y^3} \frac{dy}{dx}$$

$$= -\frac{1}{y^2} \frac{dy}{dx}$$

$$\therefore \text{②} \Rightarrow -\frac{dv}{dx} + v = x$$

$$\Rightarrow \frac{dv}{dx} - v = -x$$
 ——— ③

Here,

$$p(x) = -1$$

$$IF = e^{\int -1 dx}$$

$$= e^{-x}$$

IF $x \Rightarrow$

$$\frac{d}{dx} [e^{-x} \cdot v] = -x e^{-x}$$

$$\Rightarrow \int d[v e^{-x}] = \int -x e^{-x} dx$$

$$\Rightarrow v e^{-x} = -x \int e^{-x} dx - \int \left[\frac{d}{dx} (-x) \int e^{-x} dx \right] dx$$

$$\Rightarrow v e^{-x} = -x e^{-x} + e^{-x}$$

$$\Rightarrow \frac{e^{-x}}{y} = -x e^{-x} + e^{-x} + C \quad (\text{Ans:})$$

Exp Solve the DE: $\frac{dy}{dx} + \frac{1-2}{x} y = x^2 y^3$

Soln: $\frac{dy}{dx} + \frac{2}{x} y = x^2 y^3$

$$\Rightarrow \frac{1}{y^3} \frac{dy}{dx} + \frac{2}{x} \frac{y}{y^3} = x^2$$

$$\Rightarrow \frac{1}{y^3} \frac{dy}{dx} + \frac{2}{x} y^{-2} = x^2$$

Let, $v = y^{-2}$

$$\frac{dv}{dx} = -2 y^{-3} \frac{dy}{dx}$$

$$= -2 y^{-3} \frac{dy}{dx}$$

$$= -\frac{2}{y^3} \frac{dy}{dx}$$

$$\therefore -\frac{1}{2} \frac{dy}{dx} + \frac{2}{x} y = x^2 \quad \text{--- (ii)}$$

$$\Rightarrow \frac{dy}{dx} - \frac{4}{x} y = 2x^2 \quad \text{--- (iii)}$$

$$\Rightarrow \frac{dy}{dx} - \frac{4}{x} y = -2x^2$$

$$\text{IF} = e^{\int -\frac{4}{x} dx} = e^{-4 \ln x} = e^{\ln x^{-4}} = x^{-4} = \frac{1}{x^4}$$

$$= e^{-4 \ln x} \quad \text{--- (i)}$$

$$= e^{\ln x^{-4}}$$

$$= \frac{1}{x^4}$$

$$\Rightarrow \text{IF} \times \text{--- (iii)} \Rightarrow$$

$$\frac{1}{x^4} \frac{d}{dx} \left[\frac{1}{x^4} \cdot y \right] = -2x^2 \cdot \frac{1}{x^4}$$

$$\Rightarrow \int d \left[y \frac{1}{x^4} \right] = \int -\frac{2x^2}{x^4} dx$$

$$\Rightarrow y \cdot \frac{1}{x^4} = -2 \int x^{-2} dx$$

$$\Rightarrow y \cdot \frac{1}{x^4} = -2 \cdot \frac{x^{-2+1}}{-2+1} = -2 \frac{x^{-1}}{-1} = \frac{2}{x}$$

$$\frac{1}{y^2} \frac{1}{x^4} y + \frac{y}{x^4} = \frac{2}{x} \quad \text{--- (Ans.)}$$

Higher Order Linear DE

The general n^{th} Order Linear DE:

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1 \frac{dy}{dx} + a_0 y = f(x)$$

$$\Rightarrow y \left(a_n \frac{d^n}{dx^n} + a_{n-1} \frac{d^{n-1}}{dx^{n-1}} + \dots + a_1 \frac{d}{dx} + a_0 \right) = f(x)$$

$$\Rightarrow y P(D) = f(x) \quad \text{--- (i)}$$

If,

$$f(x) = 0 \longrightarrow \text{Homogeneous DE}$$

$$f(x) \neq 0 \longrightarrow \text{Non Homogeneous DE}$$

$$\Rightarrow \text{Homogeneous DE: } P(D)y = 0$$

$$P(D)y = 0 \quad \text{--- (ii)}$$

Solution: $y_c = \text{complementary Soln}$

$$\Rightarrow \text{Non Homogeneous DE: } P(D)y = f(x)$$

$$P(D)y = f(x) \quad \text{--- (iii)}$$

Solution: $y_p = \text{Particular Solution}$

$$\text{General Soln: } y = y_c + y_p$$

⇒ 2nd Order Homogeneous DE :

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0 \quad \text{--- (i)}$$

Soln^m $a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0$ (ii) case ⇒

$$\Rightarrow \left[y \left(a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y \right) \right] = 0$$

Let a trial solution of (i) be (iii) case ⇒

$$y = e^{mx}; \quad m \text{ is a constant}$$

From (i) we get,

$$a_0 m^2 e^{mx} + a_1 m e^{mx} + a_2 e^{mx} = 0$$

$$\Rightarrow e^{mx} (a_0 m^2 + a_1 m + a_2) = 0$$

$$\therefore a_0 m^2 + a_1 m + a_2 = 0$$

$$m = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_0 a_2}}{2a_0}$$

$$\Rightarrow m = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_0 a_2}}{2a_0}$$

$$\Rightarrow 2 \text{ roots } (m_1, m_2)$$

⇒ Case (i) m_1 and m_2 Real and Unequal $m_1 \neq m_2$

$$(i) \quad y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

⇒ Case (ii) m_1 and m_2 Real and Equal $m_1 = m_2$

$$y = c_1 e^{m_1 x} + c_2 x e^{m_1 x} \text{ [repeat]}$$

⇒ Case (iii) m_1 and m_2 Complex and Conjugate $a \pm ib$

$$y = e^{ax} [c_1 \sin bx + c_2 \cos bx]$$

Exp : $\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 2y = 0$

Soln : $\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 2y = 0$ (1)

⇒ Let,

$$y = e^{mx} \text{ --- trial solution}$$

$$\Rightarrow m^2 e^{mx} + 3m e^{mx} + 2e^{mx} = 0$$

$$\Rightarrow e^{mx} (m^2 + 3m + 2) = 0$$

↓ auxiliary eqⁿ

$$\Rightarrow 1 = m^2 + 3m + 2 = 0$$

$$\Rightarrow m = -2, m = -1 \quad [\text{Real, Unequal}]$$

$$\therefore y = c_1 e^{-x} + c_2 e^{-2x} \quad (\text{Ans.})$$

Exp

$$\frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 9y = 0$$

Soln:

$$\frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 9y = 0 \quad \text{--- ①}$$

$\Rightarrow$ Let,

$$y = e^{mx} \quad (\text{be a trial solution})$$

$$m^2 e^{mx} + [6m e^{mx} + 9 e^{mx}] = 0$$

$$\Rightarrow e^{mx} (m^2 + 6m + 9) = 0$$

$$\therefore m^2 + 6m + 9 = 0$$

$$\therefore m = -3, -3 \quad [\text{Real, Equal}]$$

$$y = c_1 e^{-3x} + c_2 x e^{-3x} \quad (\text{Ans.})$$

Exp

$$\frac{d^2y}{dx^2} + 2 \frac{dy}{dx} + 2y = 0 \quad \text{--- (i)} \quad \begin{aligned} y'(0) &= 1 \\ y(0) &= 0 \end{aligned}$$

Soln: Let $y = e^{mx}$ be a trial solution

$$\Rightarrow m^2 e^{mx} + 2m e^{mx} + 2e^{mx} = 0$$

$$\Rightarrow e^{mx} (m^2 + 2m + 2) = 0$$

$$\Rightarrow \left. \begin{aligned} m_1 &= -1 + i \\ m_2 &= -1 - i \end{aligned} \right\} \text{Complex } a \pm ib$$

Now, $y = e^{-x} (c_1 \cos x + c_2 \sin x)$ --- (ii)

$$y(0) = 0 \quad [x=0, y=0]$$

$$\therefore 0 = e^0 (c_1 \cos 0 + c_2 \sin 0)$$

$$0 = c_1$$

$$\therefore c_1 = 0$$

Again,

$$\frac{dy}{dx} = e^{-x} (c_1 \cos x + c_2 \sin x)$$

$$\Rightarrow 1 = c_2 \frac{d}{dx} [e^{-x} \sin x] \quad (c_1 = 0)$$

$$= c_2 [e^{-x} \cos x - e^{-x} \sin x]$$

$$= c_2 [e^0 \cos 0 - e^0 \sin 0]$$

$$\therefore c_2 = 1$$

$$\therefore y = e^{-x} (\sin x) \quad (\text{Ans.})$$

Lee-06

RHR

22/08/26

* Particular solution: (Non Homogeneous)

We use the method of undetermined coefficients to solve the Non homogeneous DE. Then the general

Solution is $\Rightarrow y = y_c + y_p$
 y_c $\rightarrow$ Complementary solution
 y_p $\rightarrow$ Particular solution

If, $P(D)y = f(x)$ is eqn - (1)

Then eqn - (1) is Non homogeneous DE.

The Function $f(x)$ can be constant, polynomial, exponential, sine or cosine, or finite sums and products of these function.

Trial Particular solution (y_p)

$f(x)$	y_p
$\Rightarrow$ Any Constant $\rightarrow$	$y_p = A$
$\Rightarrow ax + b \rightarrow$	$y_p = Ax + B$
$\Rightarrow ax^2 + bx + c \rightarrow$	$y_p = Ax^2 + Bx + C$
$\Rightarrow$ Polynomial of degree $n \rightarrow$	$y_p = \text{Polynomial of degree } n$
$\Rightarrow e^{ax} \rightarrow$	$y_p = Ae^{ax}$
$\Rightarrow \sin ax / \cos ax \rightarrow$	$y_p = A \cos ax + B \sin ax$

$\Rightarrow (ax+b)e^{\alpha x} \longrightarrow$	$y_p = (Ax+B)e^{\alpha x}$
$\Rightarrow e^{\alpha x} \sin \beta x \longrightarrow$	$y_p = A e^{\alpha x} \sin \beta x + B e^{\alpha x} \cos \beta x$
$\Rightarrow (ax^2) \sin \beta x \longrightarrow$	$y_p = (Ax^2+Bx+C) \sin \beta x + (Dx^2+Ex+F) \cos \beta x$
$\Rightarrow x e^{\alpha x} \cos \beta x \longrightarrow$	$y_p = (Ax+B)e^{\alpha x} \sin \beta x + (Cx+D)e^{\alpha x} \cos \beta x$

Lec-07

RHR
27/08/26Exp-01Solve the DE $y'' + 4y' - 2y = 2x^2 - 3x + 6$ $y(0)=0$
By Undetermined coefficient method $y'(0)=1$ Soln:

$$y'' + 4y' - 2y = 2x^2 - 3x + 6 \quad \text{--- (i)}$$

Let,

$$y'' + 4y' - 2y = 0 \quad \text{--- (ii)}$$

Let,

 $y = e^{mx}$ be a trial solution.① $\Rightarrow$

$$m^2 e^{mx} + 4m e^{mx} - 2e^{mx} = 0$$

$$\Rightarrow e^{mx} (m^2 + 4m - 2) = 0$$

$$\Rightarrow m^2 + 4m - 2 = 0$$

$$\therefore m_1 = -2 + \sqrt{6}$$

$$m_2 = -2 - \sqrt{6}$$

$$y_c = c_1 e^{(-2+\sqrt{6})x} + c_2 e^{(-2-\sqrt{6})x}$$

Let,

$$y_p = Ax^2 + Bx + C \text{ be a trial soln of (i)}$$

$$\therefore y_p' = 2Ax + B$$

$$y_p'' = 2A$$

$$\textcircled{1} \Rightarrow y'' + 4y' - 2y$$

$$\Rightarrow 2A + 4(2Ax + B) = 2(Ax^2 + Bx + C) = 2x^2 - 3x + 6$$

$$\Rightarrow 2A + 8Ax + 4B - 2Ax^2 - 2Bx - 2C = 2x^2 - 3x + 6$$

$$\Rightarrow x^2(-2A) + x(8A - 2B) + (2A - 4B - 2C) = 2x^2 - 3x + 6$$

$\therefore$ Equating the coefficients of $x^2, x, \text{ constant form}$

$$-2A = 2 \quad \therefore A = -1$$

$$8A - 2B = -3 \Rightarrow -8 - 2B = -3 \Rightarrow -2B = 5 \Rightarrow B = -\frac{5}{2}$$

$$2A - 4B - 2C = 6 \Rightarrow -2 - 10 - 2C = 6 \Rightarrow -2C = 18 \Rightarrow C = -9$$

$$\therefore y_p = -x^2 - \frac{5}{2}x - 9$$

General soln of eqn $\textcircled{1}$ is,

$$y = y_c + y_p$$

$$= c_1 e^{(-2+\sqrt{6})x} + c_2 e^{(-2-\sqrt{6})x} - x^2 - \frac{5}{2}x - 9$$

(Ans.)

Now,

$$y = e^{(-2+\sqrt{6})x} + e^{(-2-\sqrt{6})x} - x^2 - \frac{5}{2}x - 9$$

$$y(0) = 0 \therefore y = 0, x = 0$$

$$0 = e_1 e^{(-2+\sqrt{6}) \cdot 0} + e_2 e^{(-2-\sqrt{6}) \cdot 0} - 9$$

$$0 = e_1 + e_2 - 9 \quad 0 = y + y$$

$$e_1 = -e_2 \quad e_2 = 9 - e_1$$

$$y' = e_1 (-2+\sqrt{6}) e^{(-2+\sqrt{6})x} + e_2 (-2-\sqrt{6}) e^{(-2-\sqrt{6})x} - 2x - \frac{5}{2}$$

$$y'(0) = 1 \therefore y = 1, x = 0$$

$$1 = e_1 (-2+\sqrt{6}) e^{(-2+\sqrt{6}) \cdot 0} + e_2 (-2-\sqrt{6}) e^{(-2-\sqrt{6}) \cdot 0} - 2x - \frac{5}{2}$$

$$1 = e_1 (-2+\sqrt{6}) + e_2 (-2-\sqrt{6}) - 2x - \frac{5}{2}$$

$$1 = -2e_1 + \sqrt{6}e_1 - 2e_2 - \sqrt{6}e_2 - 2x - \frac{5}{2}$$

$$1 = -2e_1 + \sqrt{6}e_1 - 2(9 - e_1) - \sqrt{6}(9 - e_1) - 2x - \frac{5}{2}$$

$$e_1 = \frac{-2e_1 + \sqrt{6}e_1 - 18 - 9\sqrt{6} + 2e_1 + \sqrt{6}e_1 - \frac{5}{2}}{2\sqrt{6}} = \frac{-18 - 9\sqrt{6} + 2\sqrt{6}e_1}{2\sqrt{6}}$$

$$e_2 = -\frac{41 + 18\sqrt{6}}{2} + 2\sqrt{6}e_1 \quad e_1 = 8.89$$

$$e_2 = 0.11$$

$$\therefore y = \frac{8.89}{2\sqrt{6}} e^{(-2+\sqrt{6})x} + \frac{0.11}{2\sqrt{6}} e^{(-2-\sqrt{6})x} - x^2 - \frac{5}{2}x - 9$$

$$(Ans;)$$

$$+$$

$$+$$

Exp-2

Solve the DE

$$y'' + y = 4x + 10 \sin x$$

$$y(x) = 0$$

$$y'(x) = 2$$

Soln:

$$y'' + y = 4x + 10 \sin x \quad \text{--- (i)}$$

Let,

$$y'' + y = 0 \quad \text{--- (ii)}$$

Let,

$y = e^{mx}$ be a trial solution

$$\therefore m^2 e^{mx} + m e^{mx} = 0$$

$$\Rightarrow e^{mx} (m^2 + m) = 0$$

$$\Rightarrow m^2 + m = 0$$

$$\therefore m_1 = 0, m_2 = -1$$

$$\therefore y_c = e^{0x} (C_1 \sin x + C_2 \cos x) = C_1 \sin x + C_2 \cos x$$

Let,

$$y_p = (Ax + B) + (C \sin x + D \cos x)x \quad \text{[Repeat]}$$

$$y_p = (Ax + B) + Cx \sin x + Dx \cos x \quad \text{--- (iii)}$$

$$\textcircled{i} \Rightarrow y_p' = A + C[x \cos x + \sin x] + D[x \sin x + \cos x]$$

$$= A + C[\sin x + x \cos x] + D[-x \sin x + \cos x]$$

$$\textcircled{ii} \Rightarrow y_p'' = C[\cos x + x(-\sin x) + \cos x] + D[-\sin x - (x \cos x + \sin x)]$$

$$= C[2 \cos x - x \sin x] + D[-2 \sin x - x \cos x]$$

$$\textcircled{i} \Rightarrow C[2 \cos x - x \sin x] + D[-2 \sin x - x \cos x] +$$

$$(Ax + B) + (Cx \sin x + Dx \cos x) = 4x + 10 \sin x$$

$$\Rightarrow 2\cos x - \cancel{Cx\sin x} - 2D\sin x - \cancel{Dx\cos x} + Ax + B + \cancel{Cx\sin x} + \cancel{Dx\cos x} = 4x + 10\sin x$$

$$\Rightarrow 2\cos x - 2D\sin x + Ax + B = 4x + 10\sin x$$

$\therefore$ Equating coefficients from both sides:

$$x: A = 4$$

$$B = 0$$

$$C = 0$$

$$D = -5$$

(ii) $\Rightarrow$

$$y_p = 4x - 5x\cos x$$

$$\therefore y = y_p + y_c$$

$$y = (C_1 \sin x + C_2 \cos x) + 4x - 5x\cos x \quad \text{--- (iv)}$$

$$\therefore y(\pi) = 0$$

$$y'(\pi) = 2$$

$$\Rightarrow 0 = (C_1 \sin \pi + C_2 \cos \pi) + 4\pi - 5\pi \cos \pi$$

$$= -C_2 + 9\pi \therefore C_2 = 9\pi$$

$$\therefore y' = C_1 \cos x - C_2 \sin x + 4 - 5(x \sin x + \cos x)$$

$$= C_1 \cos x - C_2 \sin x + 4 + 5x \sin x - 5 \cos x$$

$$2 = C_1 \cos(\pi) - C_2 \sin(\pi) + 4 + 5\pi \sin \pi - 5 \cos \pi$$

$$2 = -c_1 - 0 + 4 + 0 + 5$$

$$\Rightarrow 2 = -c_1 + 9 \Rightarrow c_1 = 7$$

$$\therefore y = 7 \sin x + 9 \cos x + 4x - 5x \cos x \quad (\text{Ans})$$

Exp-3

Solve the DE: $y'' + 7y' + 10y = e^{-2x}$

Soln:

$$y'' + 7y' + 10y = e^{-2x}$$

Let,

$$y'' + 7y' + 10y = 0$$

Let,

$$y = e^{mx} \text{ be a trial soln}$$

$$\therefore m^2 e^{mx} + m e^{mx} + 10 e^{mx} = 0$$

$$\Rightarrow e^{mx} (m^2 + 7m + 10) = 0$$

$$\therefore m^2 + 7m + 10 = 0$$

$$\therefore m_1 = -2$$

$$m_2 = -5$$

$$y_c = c_1 e^{-2x} + c_2 e^{-5x}$$

Now,

$$e^{ax} = A e^{ax}$$

$$e^{-2x} = A e^{-2x}$$

Let,

$$y_p = (A e^{ax})^x + [\text{Repeat}]$$

$$= A x e^{ax} = A x e^{-2x}$$

$$\textcircled{i} \quad y_p' = A(x a e^{ax} + e^{ax}) = -2A x e^{-2x} + A e^{-2x}$$

$$y_p'' = A a^2 x + a e^{ax} - 2A(x - 2e^{-2x} + e^{-2x})$$

$$= 4A x + e^{-2x} - 2e^{-2x} \Rightarrow -4A x e^{-2x} - 2A e^{-2x} - 2A e^{-2x}$$

$$\textcircled{i} \Rightarrow 4A x + e^{-2x} - 2e^{-2x} + 7(-2A x e^{-2x} + A e^{-2x}) + 10(A x e^{-2x}) = e^{-2x}$$

$$\Rightarrow 4A x + e^{-2x} - 2e^{-2x} - 14A x e^{-2x} + 7A e^{-2x} + 10A x e^{-2x} = e^{-2x}$$

$$\Rightarrow x(4A - 14A e^{-2x} + 10A e^{-2x}) + 6e^{-2x} = e^{-2x}$$

$$\Rightarrow y_p' = -2A x e^{-2x} + A e^{-2x}$$

$$y_p'' = 4A x e^{-2x} - 4A e^{-2x}$$

$$\therefore 4A x e^{-2x} - 4A e^{-2x} + 7(-2A x e^{-2x} + A e^{-2x}) + 10(A x e^{-2x}) = e^{-2x}$$

$$10(A x e^{-2x}) = e^{-2x}$$

$$\Rightarrow 4A x e^{-2x} - 4A e^{-2x} - 14A x e^{-2x} + 7A e^{-2x} + 10A x e^{-2x} = e^{-2x}$$

$$10A x e^{-2x} = e^{-2x}$$

$$\Rightarrow 3A e^{-2x} = e^{-2x}$$

$$A = \frac{1}{3}$$

$$A = xy \Rightarrow f(x,y) xy$$

$$V = \int_0^1 \left[\int_0^3 f(x,y) dx \right] dy$$

$$= \iint dx dy = A$$

If the integral is

$Z = f(x,y)$ then double integral represents the Volume.

Exp-01

$$\int_0^1 \int_0^3 (1 + 4xy) dx dy$$

$$\Rightarrow \int_0^3 \left[x + 2x^2y \right]_0^3 dy$$

$$\Rightarrow \int_0^3 (1 + 2y) dy$$

$$\Rightarrow \left[y + y^2 \right]_0^3$$

$$\Rightarrow 3 + 3^2 - 0 - 0$$

$$\Rightarrow 12 - 2$$

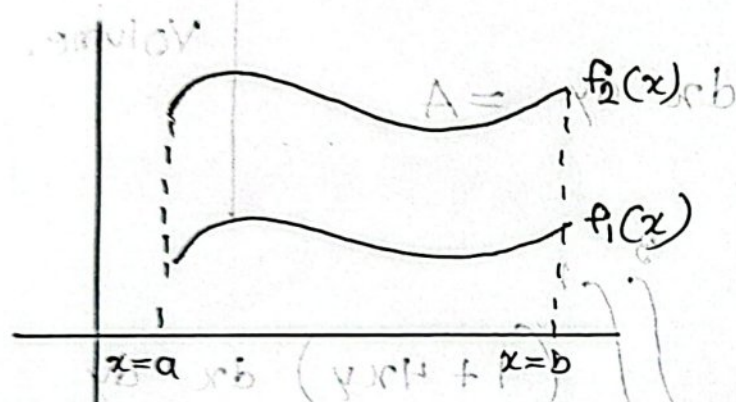
$$= 10$$

Anton [14.1] $\Rightarrow 1-16, 29-31$

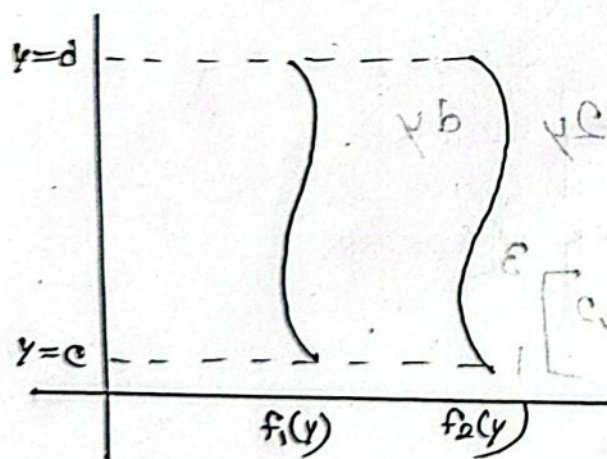
Lec- 08

RHR

Double Integral Over Non rectangular region :- V



$$V = \int_{x=a}^b \left(\int_{f_1(x)}^{f_2(x)} f(x,y) dy \right) dx$$



$$V = \int_{y=c}^d \left(\int_{f_1(y)}^{f_2(y)} f(x,y) dx \right) dy$$

Exp-01

$$\int_0^1 \int_{-x}^{x^2} xy^2 dy dx$$

Soln:

$$\int_0^1 \int_{-x}^{x^2} xy^2 dy dx$$

$$\Rightarrow \int_0^1 \left[\frac{xy^3}{3} \right]_{-x}^{x^2} dx$$

$$\Rightarrow \frac{x}{3} \int_0^1 (x^6 - (-x^3)) dx$$

$$\Rightarrow \frac{x}{3} \int_0^1 (x^6 + x^3) dx$$

$$\Rightarrow \frac{x}{3} \left[\frac{x^7}{7} + \frac{x^4}{4} \right]_0^1$$

$$\Rightarrow \frac{1}{3} \left[\frac{x^8}{8} + \frac{x^5}{5} \right]_0^1$$

$$\Rightarrow \frac{1}{3} \left[\frac{1}{8} + \frac{1}{5} \right]$$

$$= \frac{13}{120}$$

Exp-2

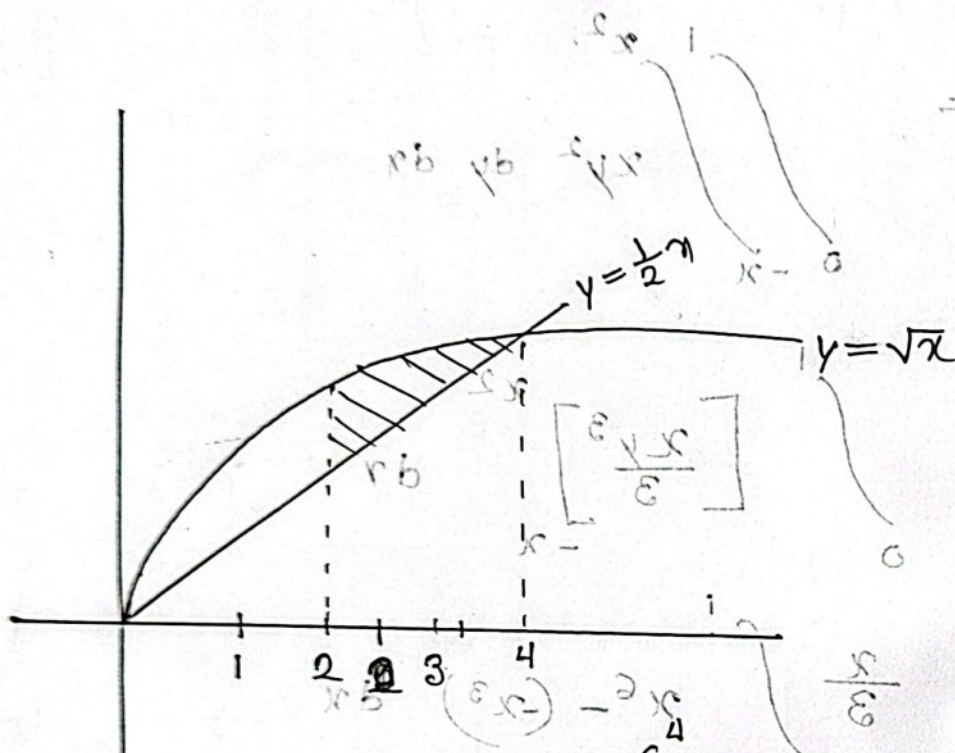
Evaluate

$\iint_R xy \, dA$ over the region bounded

by the curves

$y = \frac{1}{2}x$, $y = \sqrt{x}$, $x=2$, $x=4$

Soln:



$$\Rightarrow \int_2^4 \int_{\frac{1}{2}x}^{\sqrt{x}} xy \, dy \, dx$$

$$\Rightarrow \frac{x}{2} \int_2^4 \left[y^2 \right]_{\frac{1}{2}x}^{\sqrt{x}} dx$$

$$\Rightarrow \frac{x}{2} \int_2^4 \left(x - \frac{x^2}{4} \right) dx$$

$$\Rightarrow \frac{x}{2} \int_2^4 \frac{4x - x^2}{4} dx$$

$$\Rightarrow \frac{1}{8} \int_2^4 (4x^2 - x^3) dx$$

$$\Rightarrow \frac{1}{8} \left[4 \frac{x^3}{3} - \frac{x^4}{4} \right]_2^4$$

$$\Rightarrow \frac{1}{8} \left[\frac{64}{3} - \frac{20}{1} \right]$$

$$= \frac{11}{6} \text{ (Ans.)}$$

$$\left[\frac{1}{6} + \frac{1}{3} \right] \frac{1}{8}$$

$$\frac{1}{120}$$

Exp-3] Evaluate

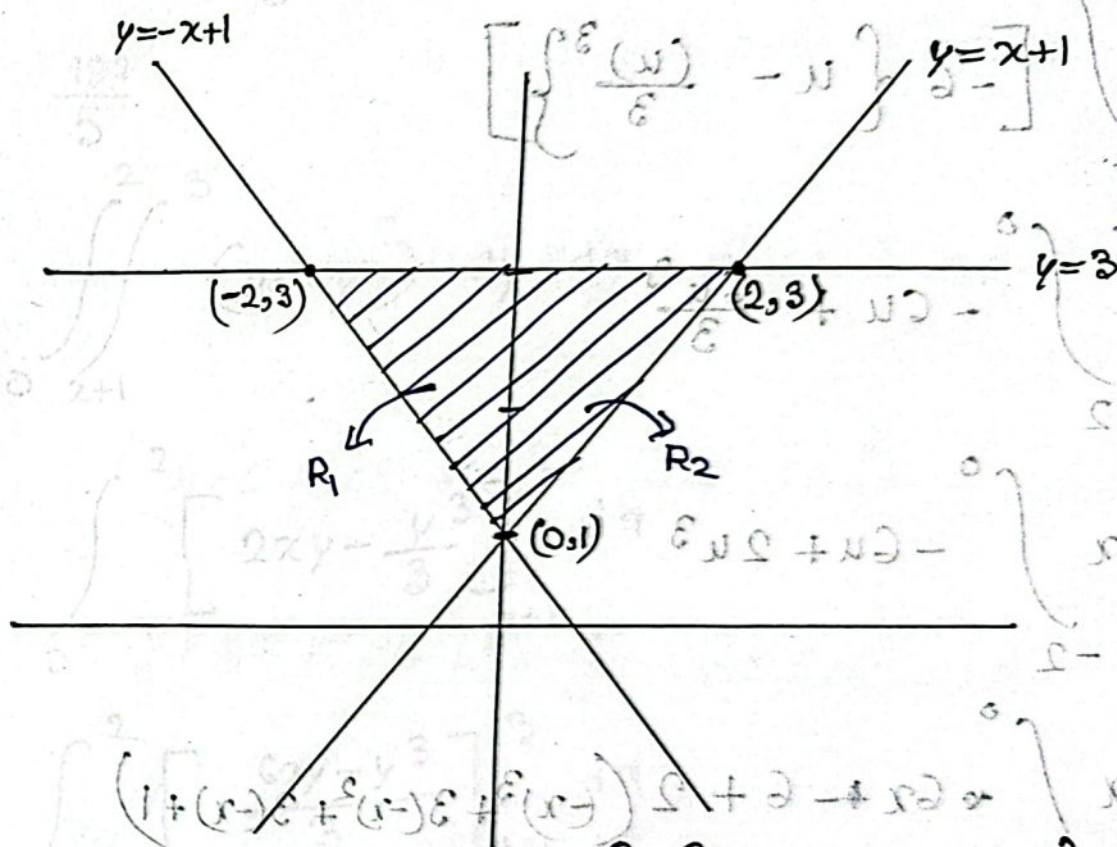
$$\iint_R (2x - y^2) dA \text{ over the}$$

triangular region

bounded by the curves $y = -x + 1$,

$$y = x + 1, y = 3$$

Soln:



$$\Rightarrow \iint_R (2x - y^2) dA = \int_{-2}^0 \int_{-x+1}^3 (2x - y^2) dy dx + \int_0^2 \int_{x+1}^3 (2x - y^2) dy dx$$

$$R_1 = \int_{-2}^0 \left[2xy - \frac{y^3}{3} \right]_{-x+1}^3 dx \Rightarrow 2x \left[-y - \frac{y^3}{3} \right]_{-x+1}^3$$

Error

Exp-04] Use a double integral to find the area of a region R enclosed between the curves $y = \frac{1}{2}x^2$ and $y = 2x$

Soln:

we know,

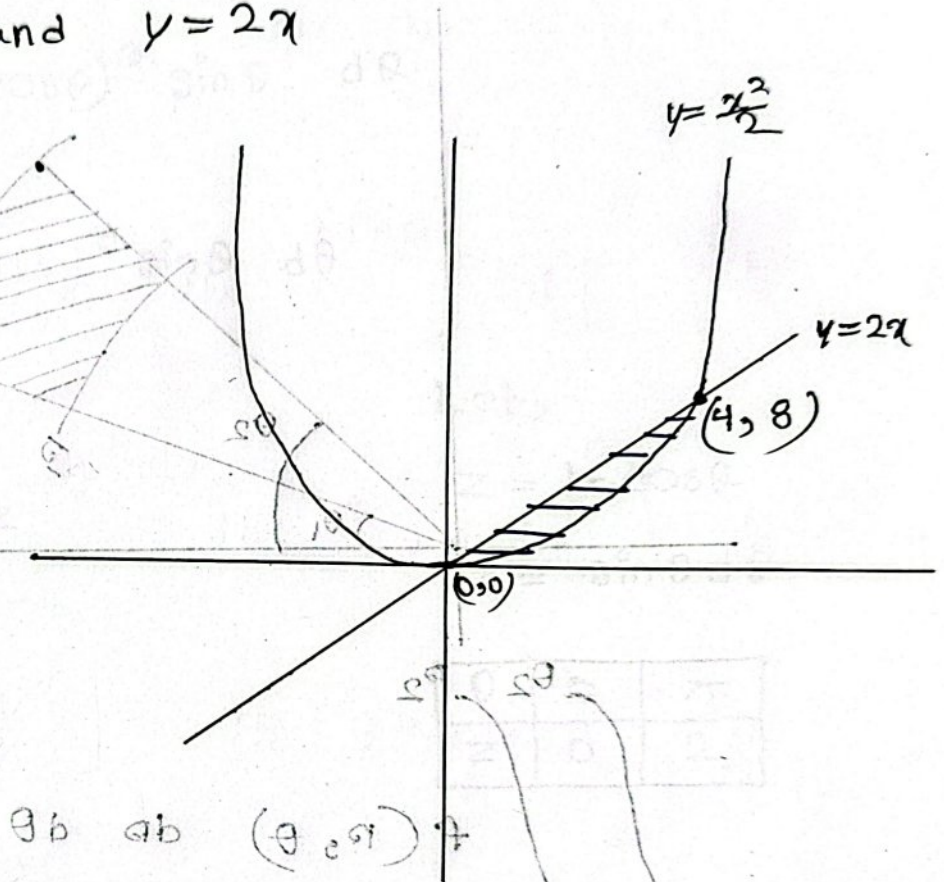
$$A = \iint_R dA$$

$$\begin{aligned} x^2 &= 4x \\ \Rightarrow x^2 - 4x + 0 &= 0 \\ \Rightarrow x(x-4) &= 0 \\ \Rightarrow x=0, x=4 \end{aligned}$$

$$\Rightarrow \int_{x=0}^4 \int_{\frac{x^2}{2}}^{2x} dy \, dx$$

$$\Rightarrow \int_0^4 \left[y \right]_{\frac{x^2}{2}}^{2x} dx$$

$$\Rightarrow \int_0^4 \left[2x - \frac{x^2}{2} \right] dx$$



$$\frac{1}{2} \int_0^4 \left[4x - \frac{x^2}{2} \right] dx$$

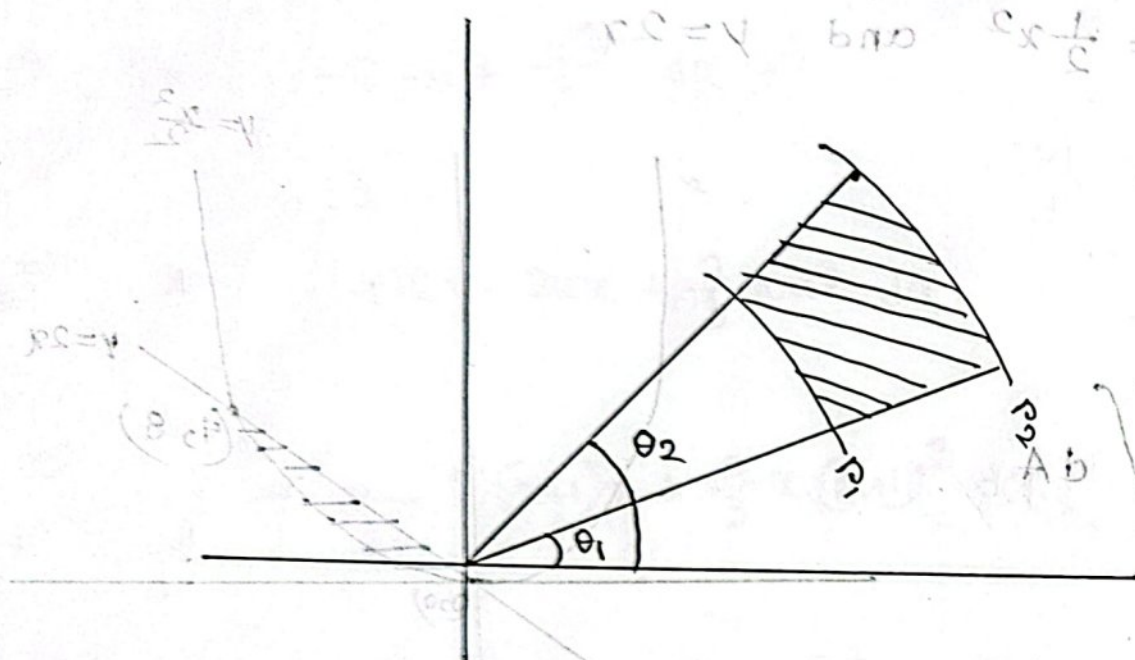
$$\Rightarrow \frac{1}{2} \left[4 \frac{x^2}{2} - \frac{x^3}{3} \right]_0^4$$

$$\Rightarrow \frac{1}{2} \left[2x^2 - \frac{x^3}{3} \right]_0^4$$

$$\Rightarrow \frac{64}{3} \text{ (Ans.)}$$

$$\Rightarrow \frac{16}{3} \text{ (Ans.)}$$

Double Integral in Polar coordinates



$$f(r, \theta) dr d\theta$$

Exp-01

$$\int_0^{\pi} \int_0^{a(1-\cos\theta)} r^2 \sin\theta dr d\theta$$

Soln:

$$\int_0^{\pi} \int_0^{a(1-\cos\theta)} r^2 \sin\theta dr d\theta$$

$$\Rightarrow \frac{1}{3} \int_0^{\pi} [a^3 \sin \theta] a (1 - \cos \theta) d\theta$$

$$\Rightarrow \frac{1}{3} \int_0^{\pi} a^3 (1 - \cos \theta)^3 \sin \theta d\theta$$

$$\Rightarrow \frac{a^3}{3} \int_0^{\pi} (1 - \cos \theta)^3 \sin \theta d\theta$$

Let,

$$z = 1 - \cos \theta$$

$$dz = \sin \theta d\theta$$

$$\Rightarrow \frac{a^3}{3} \int_0^2 z^3 dz$$

$$\Rightarrow \frac{a^3}{3} \left[\frac{z^4}{4} \right]_0^2$$

θ	0	π
z	0	2

$$\Rightarrow \frac{a^3}{3} = \frac{16a^3}{4}$$

$$\Rightarrow \frac{4a^3}{3} \text{ (Ans.)}$$

H.W

①

$$\int_0^\pi \int_0^{a \sin \theta} p \, dp \, d\theta$$

$$\Rightarrow \frac{1}{2} \int_0^\pi [p^2]_0^{a \sin \theta} d\theta$$

$$\Rightarrow \frac{1}{2} \int_0^\pi a^2 (\sin \theta)^2 d\theta$$

$$\Rightarrow \frac{a^2}{2} \int_0^\pi \sin^2 \theta d\theta$$

$$\Rightarrow \frac{a^2}{2} \int_0^\pi \frac{1 - \cos(2\theta)}{2} d\theta$$

Let,

$$\sin \theta = x$$

$$x = \sin \theta$$

$$\Rightarrow \frac{a^2}{2} \int_0^\pi \left(\frac{1}{2} - \frac{\cos(2\theta)}{2} \right) d\theta$$

$$\left(\frac{1}{2} - \frac{\cos(2\theta)}{2} \right)$$

$$\Rightarrow \frac{a^2}{2} \left[\frac{1}{2} \theta - \frac{\sin(2\theta)}{4} \right]_0^\pi$$

$$\Rightarrow \frac{a^2}{2} \times 90 \Rightarrow 45a^2$$

ii)

$$\int_0^\pi a(1+\cos\theta) \sin\theta \, d\theta$$

Soln:

$$\frac{1}{2} \int_0^\pi [a^2 \sin^2\theta] \, d\theta$$

$$\frac{1}{2} \int_0^\pi a^2 (1+\cos\theta)^2 \sin\theta \, d\theta$$

$$\frac{a^2}{2} \int_0^\pi (1 + 2\cos\theta + \cos^2\theta) \sin\theta \, d\theta$$

$$\frac{a^2}{2} \int_2^0 z^2 (-dz) = \frac{a^2}{2} \int_0^2 z^2 dz$$

$$-\frac{a^2}{2} \left[\frac{z^3}{3} \right]_2^0$$

$$-\frac{a^2}{6} \left[0 - \frac{8}{3} \right] = \frac{4a^2}{3}$$

θ	0	π
z	2	0

Lee-09

Cardioid :

$$\Rightarrow r = a(1 + \cos \theta)$$

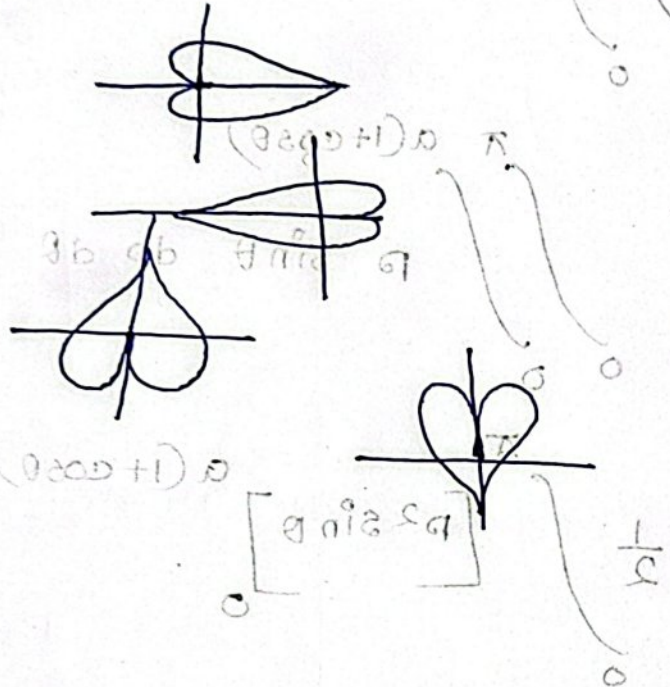
$$\Rightarrow r = a(1 - \cos \theta)$$

$$\Rightarrow r = a(1 + \sin \theta)$$

$$\Rightarrow r = a(1 - \sin \theta)$$

* r = Radial distance

* a = Constant number



Exp-01

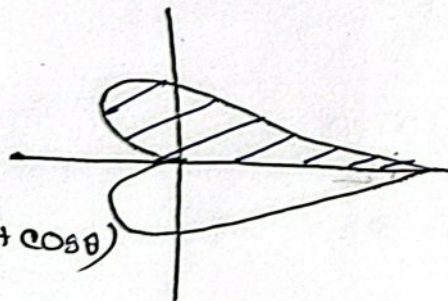
Evaluate

$$\int_R \int_0^{2\pi} r \sin \theta \, dr \, d\theta \text{ over the area}$$

of the cardioid $r = a(1 + \cos \theta)$ above the initial line.

Soln: =

Given =



$\theta = 0$ (Initial line)

r	a	$a(1 + \cos \theta)$
θ	0	π

$\Rightarrow$

$$r \sin \theta$$

$$dr \, d\theta$$

$$a$$

$$\left[\frac{r^2}{2} \right]_0^{a(1 + \cos \theta)}$$

$$\frac{a^2}{2} (1 + \cos \theta)^2$$

$$\frac{a^2}{2} (1 + \cos \theta)^2$$

$$\frac{a^2}{2} (1 + \cos \theta)^2$$

$$\frac{a^2}{2} (1 + \cos \theta)^2$$

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$$\frac{a^2}{2} (1 + \cos \theta)^2$$

$$\frac{a^2}{2} (1 + \cos \theta)^2$$

$$\frac{a^2}{2} (1 + \cos \theta)^2$$

$$\begin{aligned}
 &\Rightarrow \int_0^\pi \left[\frac{r^2}{2} \right]_0^{a(1+\cos\theta)} \sin\theta \, d\theta \\
 &\Rightarrow \frac{1}{2} \int_0^\pi a^2 (1+\cos\theta)^2 \sin\theta \, d\theta \\
 &\Rightarrow -\frac{1}{2} a^2 \int_2^0 z^2 \, dz \quad \left| \begin{array}{l} z = \sin\theta + \cos\theta \\ dz = -\sin\theta \, d\theta \end{array} \right. \\
 &\Rightarrow -\frac{a^2}{2} \left(-\frac{8}{3} \right) \\
 &\Rightarrow \frac{4a^2}{3}
 \end{aligned}$$

θ	0	π
z	2	0

* Change of Variables in Double Integrals (Cartesian to Polar)

To transform Cartesian coordinates (x, y) into polar coordinate (r, θ)

We define,

$$x = r \cos\theta$$

$$y = r \sin\theta$$

$$r = \sqrt{x^2 + y^2}$$

$$dx \, dy = J \, dr \, d\theta$$

→ Jacobian

$$\begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r(\cos^2 \theta + \sin^2 \theta) = r$$

$$\Rightarrow \int \int_R f(x, y) dx dy = \int \int_{\theta, r} f(r \cos \theta, r \sin \theta) r dr d\theta$$

$$\Rightarrow dx dy = r dr d\theta$$

Hence, the double integral in polar coordinates is given by

$$\iint_R f(x, y) dx dy = \int \int_{\theta, r} f(r \cos \theta, r \sin \theta) r dr d\theta$$

We define

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$x^2 + y^2 = r^2$$

or

Exp-01 Evaluate $\iint_R p \cos \theta \, dA$ Over the polar

region $0 \leq \theta \leq \pi$ and $0 \leq p \leq 1 - \sin \theta$

Soln:

$$\int_0^\pi \int_0^{1-\sin \theta} p \cos \theta \, p \, dp \, d\theta$$

$$\Rightarrow \int_0^\pi \left[\frac{p^2}{2} \cos \theta \right]_0^{1-\sin \theta} d\theta = \frac{1}{2} \int_0^\pi (1-\sin \theta)^2 \cos \theta \, d\theta$$

$$\Rightarrow \frac{1}{2} \int_0^\pi (1 - 2\sin \theta + \sin^2 \theta) \cos \theta \, d\theta$$

$$\Rightarrow \frac{1}{2} \int_0^\pi (1 - 2\sin \theta + \sin^2 \theta) \cos \theta \, d\theta$$

$$\Rightarrow \frac{1}{2} \int_1^{-1} z^3 \, dz$$

$$\Rightarrow 0$$

Let,

$$z = 1 - \sin \theta$$

$$dz = -\cos \theta \, d\theta$$

θ	0	π
z	1	-1

Exp-02

Evaluate

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy dx$$

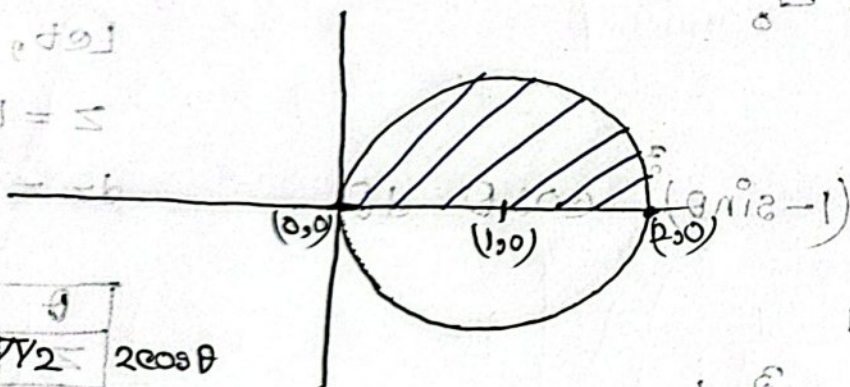
to polar coordinates. $0 \leq x \leq 2$ and $0 \leq \theta \leq \pi/2$

Soln:

$$x=0, x=2$$

$$y=0, y=\sqrt{2x-x^2}$$

$$\begin{aligned} y^2 &= 2x - x^2 \\ x^2 + y^2 &= 2x \\ x^2 - 2x + y^2 &= 0 \end{aligned} \quad \left| \quad \begin{aligned} x^2 - 2x + 1 - 1 + y^2 &= 0 \\ (x-1)^2 + y^2 &= 1 \\ (1,0) \text{ center} \\ \text{radius} &\Rightarrow r=1 \end{aligned} \right| \quad \begin{aligned} y^2 + x^2 &= 2x \\ r^2 &= 2r \cos \theta \\ r &= 2 \cos \theta \end{aligned}$$



θ	ϕ	θ
0	$\pi/2$	$\pi/2$

$$\int_{\theta=0}^{\pi/2} \int_{r=0}^{2\cos\theta} \frac{r \cos \theta}{\sqrt{r^2}} r dr d\theta$$

$$\Rightarrow \int_0^{\pi/2} \int_0^{2\cos\theta} r \cos \theta dr d\theta$$

$$\Rightarrow \int_0^{\pi/2} \int_0^{2\cos\theta} r \, dr \, d\theta$$

$$\Rightarrow \int_0^{\pi/2} \cos\theta \left[\frac{r^2}{2} \right]_0^{2\cos\theta} d\theta$$

$$\Rightarrow 2 \int_0^{\pi/2} \cos^3\theta \, d\theta$$

$$\Rightarrow 2 \int_0^{\pi/2} \frac{\cos 3\theta + 3\cos\theta}{4} d\theta$$

$$\Rightarrow \frac{1}{2} \int_0^{\pi/2} (\cos 3\theta + 3\cos\theta) d\theta$$

$$\Rightarrow \frac{1}{2} \left[\int_0^{\pi/2} \cos(3\theta) d\theta + 3 \int_0^{\pi/2} \cos\theta d\theta \right]$$

$$\Rightarrow \frac{1}{2} \left[\left[\frac{\sin 3\theta}{3} \right]_0^{\pi/2} + 3 \left[\sin\theta \right]_0^{\pi/2} \right]$$

$$\Rightarrow \left(\frac{1}{2} \right) \left[-\frac{1}{3} + 3 \right] = \frac{4}{3}$$

Exp-3

Evaluate

$$\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2+y^2)^{3/2} dy dx \quad \text{by changing}$$

to polar coordinates.

Soln:

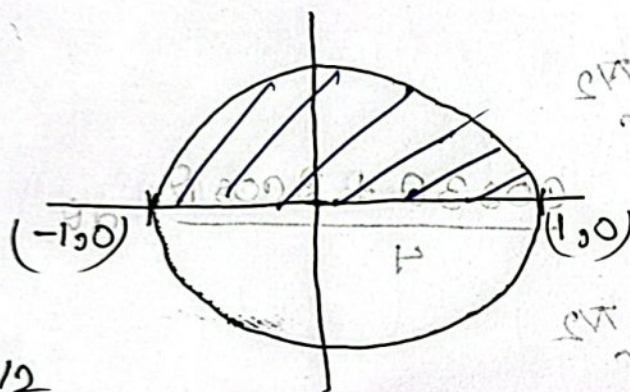
$$x=1, \quad x=-1$$

$$y=0, \quad y=\sqrt{1-x^2}$$

$$y^2+x^2=1$$

$$\Rightarrow r^2=1^2$$

$$r=1$$



$$\Rightarrow \int_0^{\pi} \int_0^1 (r^2)^{3/2} r dr d\theta$$

$$\Rightarrow \int_0^{\pi} \int_0^1 r^4 dr d\theta$$

$$\Rightarrow \int_0^{\pi} \left[\frac{r^5}{5} \right]_0^1 d\theta = \int_0^{\pi} \frac{1}{5} d\theta = \left[\frac{\theta}{5} \right]_0^{\pi} = \frac{\pi}{5} \quad (\text{Ans})$$

Exp-04

Find

$$\iint_R \sin \theta \, dA, \text{ when } R \text{ is the region}$$

in the 1st Quadrant that is outside the circle $p=2$ and inside the cardioid $p=2(1+\cos \theta)$

Soln:

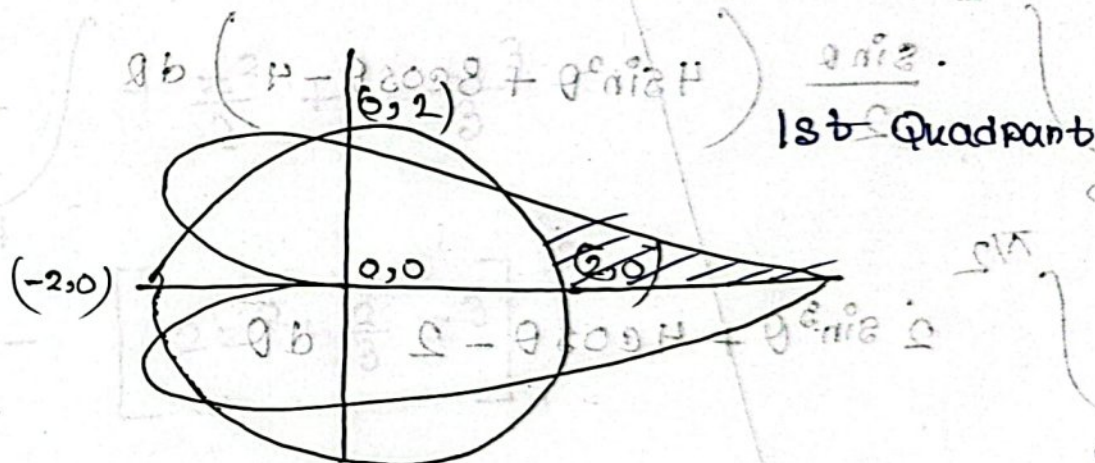
$$p = 2(1 + \cos \theta)$$

$$a = 2$$

$$x^2 + y^2 = 4$$

$$x^2 + y^2 = 2^2$$

center $(0,0)$ $p=2$



$$\Rightarrow \int_{\theta=0}^{\pi/2} \int_{p=2}^{2(1+\cos \theta)} \sin \theta \, p \, dp \, d\theta$$

$$\Rightarrow \int_{\theta=0}^{\pi/2} \left[\frac{p^2}{2} \right]_2^{2(1+\cos \theta)} \sin \theta \, d\theta$$

$$\Rightarrow \int_0^{\pi/2} \frac{\sin \theta}{2} \left[(2(1+\cos \theta))^2 - 2^2 \right] d\theta$$

$$\Rightarrow \int_0^{\pi/2} \frac{\sin \theta}{2} \left[4(1+2\cos \theta + \cos^2 \theta) - 4 \right] d\theta$$

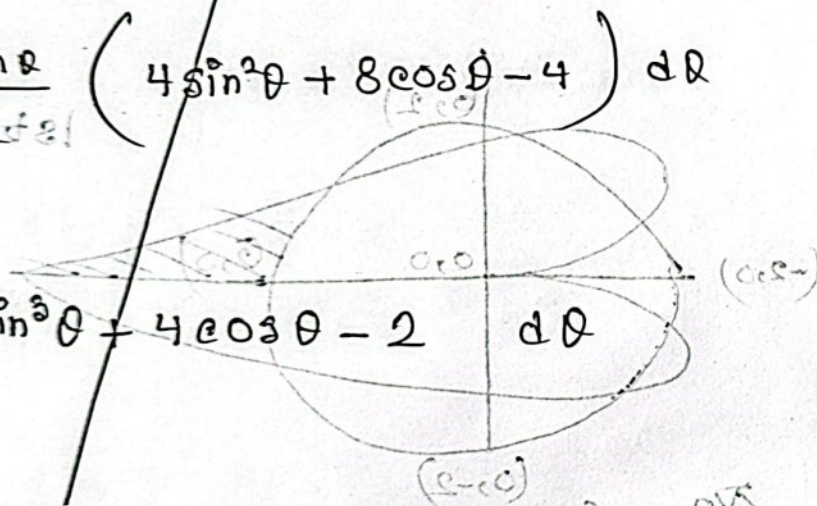
$$\Rightarrow \int_0^{\pi/2} \frac{\sin \theta}{2} \left[4\sin^2 \theta + 8\cos \theta - 4 \right] d\theta$$

$$\Rightarrow \int_0^{\pi/2} \sin \theta (2\sin^2 \theta + 4\cos \theta - 2) d\theta$$

$$\Rightarrow \int_0^{\pi/2} (2\sin^3 \theta + 4\cos \theta - 2) d\theta$$

$$\Rightarrow \int_0^{\pi/2} \frac{\sin \theta}{2} (4 + 8\cos \theta + 4\cos^2 \theta - 4) d\theta$$

$$\Rightarrow \int_0^{\pi/2} (4\sin \theta \cos \theta + 2\sin \theta \cos^2 \theta) d\theta$$



Q. 10

$$z = \cos \theta$$

$$\Rightarrow dz = -\sin \theta d\theta$$

θ	0	$\pi/2$
z	1	0

$$-\int_1^0 (4z + 2z^2) dz$$

$$\Rightarrow -\int_1^0 4 \frac{z^2}{2} + 2 \frac{z^3}{3}$$

$$\Rightarrow -\left[2z^2 + \frac{2}{3} z^3 \right]_1^0$$

$$\Rightarrow -\left[-\left(2 + \frac{2}{3} \right) \right]$$

$$\Rightarrow \frac{8}{3} \text{ (Ans.)}$$

* Example Maths

Example-1

2018

$$\int_0^1 (x^2 + xy) dx$$

$$\int_0^1 (x^2 + xy) dx$$

$$\left[\frac{x^3}{3} + \frac{xy^2}{2} \right]_0^1$$

$$\left(\frac{1}{3} + \frac{y}{2} \right) - \left(0 + 0 \right)$$

$$\frac{1}{3} + \frac{y}{2}$$

$$\left[\frac{1}{3} + \frac{y}{2} \right]_0^1$$

$$\left(\frac{1}{3} + \frac{y}{2} \right)$$